Solutions of Assignment #1 part 1, Total points: 50
(Course: CS 401)

These are the first two problems for Assignment 1. **The remaining problems of Assignment 1 will be given out later.**

Problem 1 (25 points): Consider the stable matching problem as taught in class. Suppose that we have only three men, say m_1, m_2 and m_3, and only three women, say w_1, w_2 and w_3, with their corresponding preference lists. Suppose also that the matching m_1–w_1, m_2–w_2, m_3–w_3 is a stable matching. We make the following claim:

> in this case the matching m_1–w_1, m_2–w_3, m_3–w_2 can **never** be a stable matching.

Your task is to decide if our claim is true or false. For this purpose, do the following.

- Either prove the claim is indeed correct. Such a proof should work **no matter** what the preferences of the men and women are, as long as m_1–w_1, m_2–w_2, m_3–w_3 is a stable matching.
- Or, prove the claim made is wrong by giving a counter-example. The counter-example should provide the preferences lists of every man and woman, and show that for these preference lists **both** m_1–w_1, m_2–w_2, m_3–w_3 and m_1–w_1, m_2–w_3, m_3–w_2 are indeed stable matchings.

Solution: The following preference lists constitute a counter-example since both both m_1–w_1, m_2–w_2, m_3–w_3 and m_1–w_1, m_2–w_3, m_3–w_2 are stable matchings.

	first	second	third
m_1	w_1	w_2	w_3
m_2	w_2	w_3	w_1
m_3	w_3	w_2	w_1

	first	second	third
w_1	m_1	m_2	m_3
w_2	m_3	m_2	m_1
w_3	m_2	m_3	m_1

Problem 2 (25 points): Give examples of two continuous functions $f(n)$ and $g(n)$ of positive real inputs n such that $f(n) \neq O(g(n))$, $g(n) \neq O(f(n))$, $f(n) \neq \Omega(g(n))$ and $g(n) \neq \Omega(f(n))$.

> HINT: You can specifiy some of the values of $f(n)$ and/or $g(n)$ depending on some condition on n, such as when n is even or odd, but make sure that your function is continuous and defined for all positive real n.

Solution

$$f(n) = \begin{cases} n^2, & \text{if } n \text{ is an even positive integer} \\ n^4, & \text{if } n \text{ is an odd positive integer} \\ f\left(\lfloor n \rfloor\right) + \Big(f\left(\lceil n \rceil\right) - f\left(\lfloor n \rfloor\right)\Big)\Big(n - \lfloor n \rfloor\Big), & \text{otherwise} \end{cases}$$

$f(n)$ is a continuous function of n since the function $h(n) = f\left(\lfloor n \rfloor\right) + \Big(f\left(\lceil n \rceil\right) - f\left(\lfloor n \rfloor\right)\Big)\Big(n - \lfloor n \rfloor\Big)$ satisfy $h(\lfloor n \rfloor) = f(\lfloor n \rfloor)$ and $h(\lceil n \rceil) = f(\lceil n \rceil)$. The function $g(n)$ can similarly be defined as

$$g(n) = \begin{cases} n^4, & \text{if } n \text{ is an even positive integer} \\ n^2, & \text{if } n \text{ is an odd positive integer} \\ g\left(\lfloor n \rfloor\right) + \Big(g\left(\lceil n \rceil\right) - g\left(\lfloor n \rfloor\right)\Big)\Big(n - \lfloor n \rfloor\Big), & \text{otherwise} \end{cases}$$

and a similar argument shows that $g(n)$ is also a continuous function of n.

Proof of $g(n) \neq O\big(f(n)\big)$ and $f(n) \neq \Omega\big(g(n)\big)$: Assume, for the sake of contradiction, that $g(n) = O\big(f(n)\big)$. This implies that there exists two positive constants n_0 and c such that $g(n) \leq cf(n)$ for all $n \geq n_0$. Let $n_1 = \max\left\{2, 2\lceil c \rceil\right\} \times n_0 \geq n_0$. Note that n_1 is an even integer and thus,

$$g\left(n_1\right) \leq c\, f\left(n_1\right) \equiv n_1^4 \leq c\, n_1^2 \equiv n_1 \leq \sqrt{c}$$

and the last inequality above contradicts the choice of n_1. The same proof also shows that $f(n) \neq \Omega\big(g(n)\big)$.

Proof of $f(n) \neq O\big(g(n)\big)$ and $g(n) \neq \Omega\big(f(n)\big)$: Assume, for the sake of contradiction, that $f(n) = O\big(g(n)\big)$. This implies that there exists two positive constants n_0 and c such that $f(n) \leq cg(n)$ for all $n \geq n_0$. Let $n_1 = \max\left\{3, 2\lceil c \rceil + 1\right\} \times n_0 \geq n_0$. Note that n_1 is an odd integer and thus,

$$f\left(n_1\right) \leq c\, g\left(n_1\right) \equiv n_1^4 \leq c\, n_1^2 \equiv n_1 \leq \sqrt{c}$$

and the last inequality above contradicts the choice of n_1. The same proof also shows that $g(n) \neq \Omega\big(f(n)\big)$.